

Escape Sequence

Game Design Document

The Space Cadets | Version 1.0

1. Game Overview

Escape Sequence is a first-person 2D card game set aboard a failing spaceship. Inspired by the tone and atmosphere of *Resident Evil 7* DLC "21" and *Mouthwashing*, the game blends survival tension with a blackjack-style card mechanic. Players take on the role of astronauts racing to reach the ship's escape pods before the vessel is lost entirely.

To reach the escape pods, players must pass through three increasingly difficult obstacles — each a round of a custom card game similar to blackjack. Survival is not guaranteed. There are only **3 escape pods**, and only players who have earned a **P.A.C. (Personal Access Card)** keycard may board one. With up to 6 players competing and only 3 keycards available, not everyone makes it out.

2. Players

2.1 Human Players

- 1 to 6 players may participate
- Players enter their name at the start of the game
- All players are astronauts competing to reach the escape pods
- Only 3 players can survive

2.2 AI Player (The System)

- There is always one built-in AI opponent that all players compete against
 - The AI grows more aggressive with each round, taking greater risks to get closer to 21
 - The AI is still cautious enough that busting is uncommon in early rounds
 - In Round 3, the AI uses any specialty cards it has been holding, making the final round the most dangerous
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3. Game Structure

3.1 The Deck

The deck consists of cards numbered **1–9**. The deck size scales with the number of players — each player adds 9 cards to the deck:

Players	Deck Size
1	9 cards
2	18 cards
3	27 cards
4	36 cards
5	45 cards
6	54 cards

The deck is randomized at the start of each round.

3.2 Specialty Cards

Specialty cards are a unique mechanic that adds strategy and unpredictability to the game. They are drawn from a separate randomized deck.

Card	Effect
Shield	Return the card you just drew — it is discarded and does not count toward your total
Wild	Choose the value of the card yourself — functions as an immediate win condition
Reverse	The drawn card is subtracted from your total instead of added
Freeze	Prevents a chosen player from hitting on their next turn
Swap	Swap one of your face-up cards with one of the AI's face-up cards
Peek	Choose to view either the AI's face-down card or the next card in the deck

Keycard as Specialty Card: If a player earns two keycards, the second keycard may be converted into a **Shield** specialty card, or gifted to another player who does not have one.

4. Round-by-Round Breakdown

Round 1 — The First Obstacle

1. Each player is dealt **2 cards face-up**
2. The AI is dealt **1 card face-down** (hidden) and **1 card face-up**
3. Players take turns choosing to **Hit** or **Stay**
 - **Hit:** Draw another card from the deck
 - **Stay:** Keep your current total and end your turn
4. Once all players have stayed, results are calculated

Outcomes:

- **Closest to 21 without busting** → Player advances and receives a P.A.C. keycard
- **Under 21 but not closest** → Player advances without a keycard
- **Over 21 (bust)** → Player is eliminated
 - *Exception:* If the AI also busts, all players advance — but only the closest player still receives the keycard
- **Tie between players** → A bonus round is played with no specialty cards; the tiebreaker winner receives the keycard

Round 2 — The Second Obstacle

- All surviving players are dealt **1 specialty card** at the start of the round
- Players may use their specialty card during this round or hold it and carry it into Round 3
- The AI receives a specialty card but **does not use it this round** — it saves it for Round 3
- Round play otherwise follows the same rules as Round 1

Round 3 — The Final Obstacle

- All surviving players are dealt **1 additional specialty card**
 - Players who held their Round 2 card may now use **both specialty cards**
 - The AI uses its held specialty card this round, making it the most difficult obstacle
 - Round play follows the same rules as Round 1
 - After results are calculated, players with a keycard may board an escape pod
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5. Win & Loss Conditions

Outcome	Result
Player has a keycard after Round 3	Player escapes the ship — Victory
Player survives all rounds but has no keycard	Player cannot board a pod — Eliminated
Player busts and AI does not	Player is dead — Eliminated
Player busts and AI also busts	Player survives the round but receives no keycard
All players are eliminated	Game over — No survivors

6. Game Flow

The following summarizes the full activity flow of the game:

1. Players log in and enter their names
 2. Players may view the demo/tutorial before starting
 3. Game begins — players cycle through Rounds 1, 2, and 3
 4. Each round: players choose to Hit, Stay, or use a Specialty Card → all players stay → results are calculated
 5. After each round, surviving players proceed; eliminated players are removed
 6. After Round 3, players with a keycard escape the ship
 7. End of game — results are recorded
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7. Features

- **Player Name Entry:** Each player enters their name before the game begins
 - **Tutorial / Demo:** A built-in demo explains the rules and can be viewed at any time during play
 - **Multiplayer Support:** 1 to 6 human players with a single AI opponent
 - **Progressive Difficulty:** The AI becomes increasingly aggressive across rounds
 - **Specialty Card System:** Strategic one-use cards that can change the outcome of a round
 - **Tie-Breaker Rounds:** Bonus rounds resolve ties with no specialty cards allowed
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8. Technical Objects (Code Structure)

The following objects form the core of the game's implementation:

Object	Description
Deck	Randomized deck of numbered cards (1–9), scales with player count
Card	Represents a single card with a value and face-up/face-down state
SpecialtyDeck	Separate randomized deck of specialty cards
Player	Stores player name, hand, keycard status, and specialty cards held
AIPlayer	Extends Player; increases risk tolerance each round; holds specialty card until Round 3
Game	Core game logic — manages rounds, turn order, result calculation, and win conditions

9. Use Cases

The following actors interact with the system:

- **Player:** Can Start the game, View Demo, Hit, Stay, Use Specialty Cards, Receive a Keycard, and Escape the Ship
- **AI Player:** Participates in Deal Cards, Calculate Result, Go to Next Round, Escape Ship, and End Game events

Key relationships include Calculate Result being triggered as part of both Start and Go to Next Round, and Escape Ship being an extension of the End Game condition.
